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Long-range electrostatics for machine learning interatomic potentials is easier than we thought ✓

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ABSTRACT

The lack of long-range electrostatics is a key limitation of modern machine learning interatomic potentials (MLIPs), hindering reliable applications to interfaces, charge-transfer reactions, polar and ionic materials, and biomolecules. In this Perspective, we distill two design principles behind the Latent Ewald Summation framework, which can capture long-range interactions, charges, and electrical response just by learning from standard energy and force training data: (i) use a Coulomb functional form with environment-dependent charges to capture electrostatic interactions, and (ii) avoid explicit training on ambiguous density functional theory partial charges. When both principles are satisfied, substantial flexibility remains: essentially any short-range MLIP can be augmented; charge equilibration schemes can be added when desired; dipoles and Born effective charges can be inferred or fine-tuned; and charge/spin-state embeddings or tensorial targets can be further incorporated. We also discuss current limitations and open challenges. Together, these minimal, physics-guided design rules suggest that incorporating long-range electrostatics into MLIPs is simpler and perhaps more broadly applicable than is commonly assumed.

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I. INTRODUCTION

There is no denying that accurate modeling of the $1/r$ -decaying long-range electrostatics is crucial in atomistic simulations, particularly for interfaces, charge-transfer processes, polar and ionic materials, and biomolecular systems.^{1–3} The open question has always been how “easy” it is to incorporate such long-range interactions into the machine learning interatomic potential (MLIP) framework.

Most MLIPs are built on the locality assumption:⁴ the energy and forces associated with a central atom are largely determined by its local neighbors.^{3,6} The total short-range energy of a system is thus approximated as a sum of local atomic energy contributions E_i from each atom i , $E^{\text{sr}} = \sum_{i=1}^N E_i$. Early MLIPs^{7,8} define the local environment using a fixed spatial cutoff radius, typically on the order of 6 Å. Later message-passing neural networks (MPNNs)^{9–14} propagate information across multiple interaction shells, effectively increasing the receptive field. However, this is no guarantee that long-range

interactions are faithfully captured, as deep MPNNs can suffer from information degradation and over-smoothing.^{15,16}

In recent years, various strategies^{6,17–52} have been proposed to combine electrostatics with MLIPs. In Fig. 1, we illustrate the representative developments and the conceptual connections between them. This is by no means exhaustive and is skewed toward the earlier methods in each category, particularly those that have inspired our own work.

The methods in the purple shaded region of Fig. 1 capture long-range effects by embedding global geometrical information directly into the model representation.^{37–42,44,45} This includes approaches based on global or reciprocal-space message passing,^{38,44} the use of virtual global nodes for communicating global information,⁴⁵ and those constructing explicit long-range descriptors.^{39–41} In these frameworks, long-range interactions are not introduced as a physical electrostatic energy term, but rather as part of the learned global feature space from which the neural network can infer nonlocal interactions.

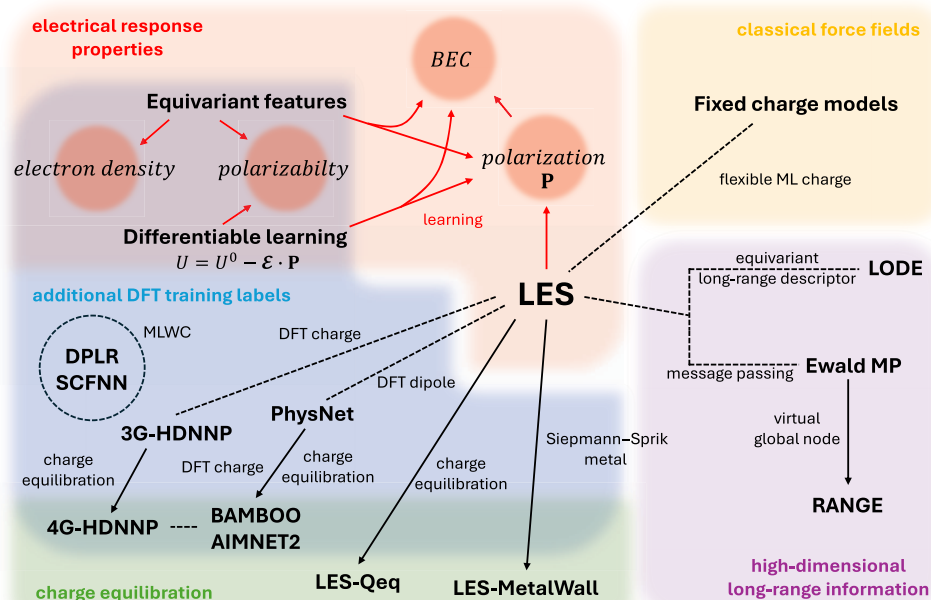


FIG. 1. Map of representative strategies for learning long-range interactions. The black solid arrows indicate an extension or variant of a method. The black dashed arrows represent conceptual connections. The red arrows indicate the learning of physical quantities. The blue shaded region indicates methods that use additional DFT training labels; the green shaded region denotes methods with charge equilibration; the purple shaded region marks the methods using high-dimensional long-range representations; the yellow shaded region represents the classical force fields with fixed charges; and the red shaded region indicates the methods that can learn electrical response properties. Abbreviations: LES, Latent Ewald Summation;²¹ HDNNP, high-dimensional neural network potential;^{23,31} BAMBOO, ByteDance AI molecular simulation booster;²⁵ AIMNET, atoms-in-molecules neural network potential;³² Ewald MP, Ewald message passing; LODE, long-distance equivariant;³⁹ RANGE, relaying attention nodes for global encoding;⁴⁵ DPLR, deep potential long-range;³⁵ SCFNN, self-consistent field neural network;³⁶ Qeq, charge equilibration scheme proposed in Refs. 47 and 53; BEC, Born effective charge tensors; Equivariant features, direct prediction of the tensorial quantities;^{62,66,67} and differentiable learning, learn corresponding response properties by taking derivatives.^{54,55} U is electric enthalpy; U^0 is the energy in the absence of the electric field; P is polarization; and $\mathcal{E}$ is a uniform electric field.⁵⁴

The methods inside the blue shaded region of Fig. 1 encode physically motivated quantities, such as atomic charges,^{22–25,27–29,31–34,48,49} typically trained on electronic information from density functional theory (DFT). Notably, 3G-HDNNP^{23,31} learns DFT partial charges,³¹ and this was later refined in 4G-HDNNP²³ that further include the charge equilibration (Qeq) scheme.^{47,53} PhysNet,²² BAMBOO,²⁵ and AIMNET2³² include DFT dipoles in the loss function and predict atomic partial charges. The deep potential long-range (DPLR) model³⁵ and the self-consistent field neural network (SCFNN)³⁶ learn from the maximally localized Wannier centers (MLWCs) for insulating systems.

In a separate vein, the methods highlighted in the red shaded region specifically target electrical response properties, such as polarization (P),^{54–61} polarizability,^{54,57,58} electron density,^{62–64} and Born effective charge (BEC) tensors (Z^*).^{54–56,59–61,65} These quantities may be learned directly as tensorial outputs using equivariant features^{55,56,62,63,65–68} or through differentiable learning frameworks in which response properties are obtained as derivatives of learned quantities,^{54,60} for example, BECs as derivatives of polarization with respect to atomic displacements^{54,56,60} and polarizations P as derivatives of energy with respect to the external electric field $\mathcal{E}$.^{54,57,58} Related developments by Grisafi *et al.*^{62,64} introduced long-range

symmetry-adapted models for predicting electron densities and the electronic density response under applied fields.

Latent Ewald Summation (LES)^{21,59–61} is at the intersection of the three aforementioned approaches: LES explicitly employs atomic charges to construct a physical long-range Coulomb electrostatic energy term; however, these charges are not learned from DFT electronic labels. Instead, they are inferred as latent, environment-dependent features solely from training on standard energy and force data. Meanwhile, electrical response properties such as P and Z^* emerge naturally.^{59–61}

The LES algorithm is simple: the local invariant features on each atom i (same as the ones for predicting local energies E_i) predict the latent atomic charge q_i^{les} via a neural network. These q_i^{les} values are then used to compute long-range electrostatic energy E^{lr} . Pairwise summation is used for finite systems,

$$E^{\text{lr}} = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{j=1}^N [1 - \varphi(r_{ij})] \frac{q_i^{\text{les}} q_j^{\text{les}}}{r_{ij}}, \quad (1)$$

where ϵ_0 is the vacuum permittivity, $\varphi(r) = \text{erfc}(\frac{r}{\sqrt{2}\sigma})$ is the complementary error function, and σ is a smearing factor typically ≈ 1 Å. It is worth noting that for moderately large molecules, it may still be

computationally advantageous to setup a periodic box with sufficient vacuum space. Ewald summation is used for periodic systems,

$$E^{\text{lr}} = \frac{1}{2\epsilon_0 V} \sum_{0 < k < k_c} \frac{1}{k^2} e^{-\sigma^2 k^2/2} |S(\mathbf{k})|^2, \quad (2)$$

with

$$S(\mathbf{k}) = \sum_{i=1}^N q_i^{\text{les}} e^{i\mathbf{k}\cdot\mathbf{r}_i}, \quad (3)$$

where $\mathbf{k}$ is the reciprocal wave vector and V is the cell volume. Finally, the electrostatic energy is added to the short-range energy to form the total potential energy of the system, $E = E^{\text{sr}} + E^{\text{lr}}$.

For a finite system, the polarization is $P = \sum_{i=1}^N q_i \mathbf{r}_i$ and the BEC for an atom i is

$$Z_{i\alpha\beta}^* = \frac{\partial P_\alpha}{\partial r_{i\beta}}, \quad (4)$$

with α and β labeling Cartesian directions. For a homogeneous periodic system,

$$Z_{i\alpha\beta}^* = \lim_{k \rightarrow 0} \Re \left[\exp(-ikr_{i\alpha}) \frac{\partial P_\alpha(k)}{\partial r_{i\beta}} \right], \quad (5)$$

where $P_\alpha(k) = \sum_{i=1}^N \sqrt{\epsilon_\infty} \frac{q_i^{\text{les}}}{ik} \exp(ikr_{i\alpha})$ and ϵ_∞ is high-frequency (electronic) relative permittivity.

In addition to the simple algorithm, the implementation of LES into any short-range MLIP is also easy. Thus far, LES has been integrated with MACE,¹² NequIP,¹¹ Allegro,⁶⁹ CACE,⁷⁰ and CHGNet,⁷¹ and these implementations are discussed and benchmarked in

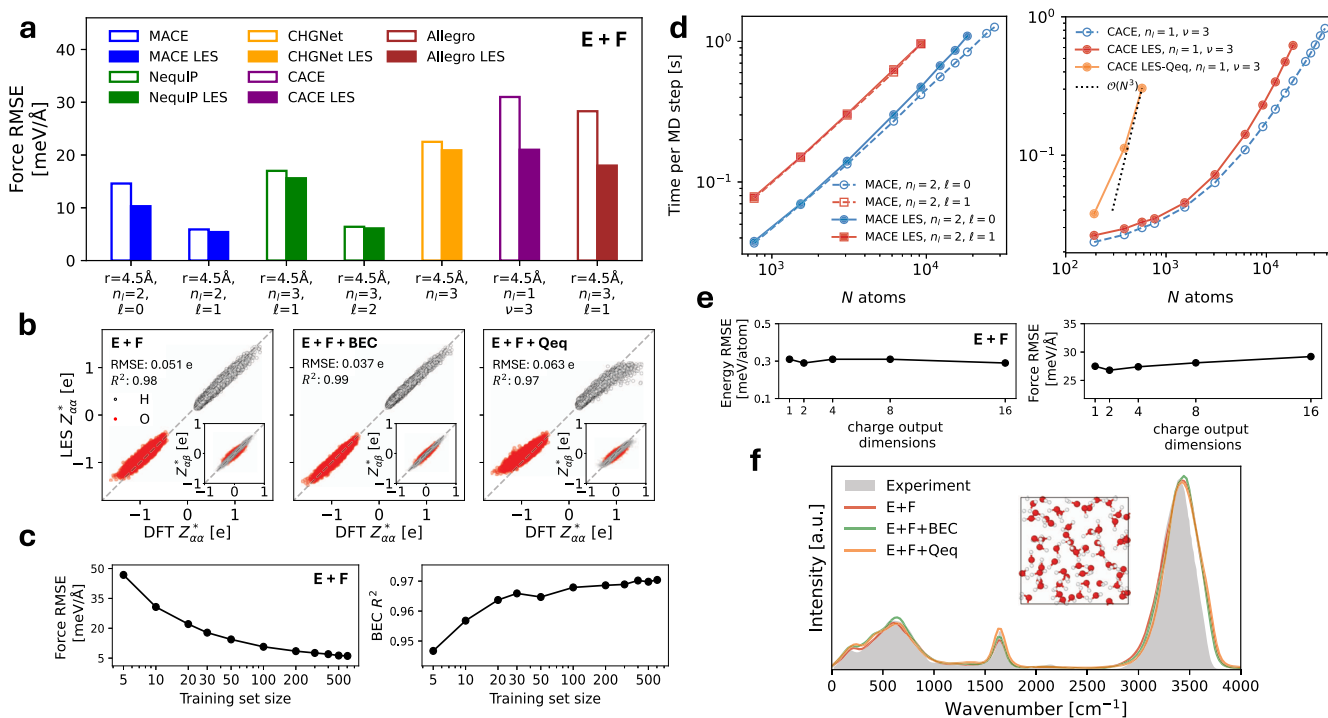


FIG. 2. Benchmark results of short-range and LES-augmented MLIPs for bulk RPBE-D3 water. Panel (a): The test force root mean square errors (RMSEs) for baseline MLIPs (hollow bars), and the corresponding LES-augmented models (solid bars) for different architectures (MACE,¹² NequIP,¹¹ CHGNet,⁷¹ CACE,⁷⁰ and Allegro⁶⁹). The cutoff r , the number of layers n_l , the order of irreducible representations (rotation order) ℓ , and body order ν are indicated for each MLIP. The results are reproduced from Ref. 61. Panel (b): Parity plots comparing the diagonal components of the BEC tensor ($Z_{\alpha\alpha}^*$) of CACE LES with RPBE-D3 reference values for 100 bulk water configurations;⁵⁵ the insets show the off-diagonal components ($Z_{\alpha\beta}^*$ with $\alpha \neq \beta$). E + F denotes a CACE LES model trained on energy and forces; the results for the model trained on E + F + BEC is reproduced from Ref. 60 and was trained on a smaller set of 100 configurations with BEC labels; and E + F + Qeq incorporates a charge equilibration (Qeq) scheme.^{47,53} The test RMSE values for forces in meV/Å are 21.0 (E + F), 25.3 (E + F + BEC), and 21.1 (E + F + Qeq). All three models use the same CACE settings ($r = 4.5 \text{ Å}$, $n_l = 1$, and $\nu = 3$). Panel (c): The test force RMSEs and BEC R^2 values of MACE LES models ($r = 4.5 \text{ Å}$, $n_l = 2$, and $\ell = 1$) as a function of training set size. Panel (d): Computational performance benchmarks of MD simulations of bulk liquid water for varying system sizes (N) using LES-augmented MLIPs in single-precision (float32) with ASE implementations⁷³ performed on an NVIDIA L40S GPU (48 GB memory). The left panel shows the timing of MD simulations using the MACE models ($r = 4.5 \text{ Å}$, $n_l = 2$) with different ℓ , both without (empty symbols and dashed lines) and with LES (filled symbols and solid lines). The right panel shows the timing of the CACE models ($r = 4.5 \text{ Å}$, $n_l = 1$, and $\nu = 3$), without LES (blue line), with LES (red line), and with Qeq scheme implementation (orange line). A reference line for N^3 scaling is included to illustrate the performance. The CACE LES model with the Qeq scheme shows cubic scaling with respect to the number of atoms. Panel (e): The test RMSEs for energy and force of CACE LES models ($r = 4.5 \text{ Å}$, $n_l = 1$, and $\nu = 2$) with varying charge output dimensions. Panel (f): Infrared (IR) absorption spectra of bulk liquid water obtained with the E + F, E + F + BEC, and E + F + Qeq CACE LES models. The experimental IR spectrum⁷⁴ is included for comparison. E + F and E + F + BEC results are taken from Ref. 60.

Ref. 61. In addition, LES has been independently implemented in TACE.⁵⁰ In Refs. 21, 59–61, and 72, LES has been extensively benchmarked on diverse systems, including bulk liquids,^{21,59–61} molecular crystals,⁶¹ polar molecules,^{21,59,61} ferroelectrics,⁶⁰ and electrochemical interfaces.^{59,72} Overall, the LES augmentation was shown to universally improve the accuracy and ensure the reliable extraction of electrical response properties, including BECs,^{59–61} dipoles for finite systems,^{59,61} infrared spectra,^{60,61} and ionic electrical conductivities.⁶⁰ Meanwhile, adding LES only introduces a marginal computational overhead,^{59,61} making it effectively a “free lunch.”

Throughout this Perspective, for illustrative purposes, we highlight representative results on the RPBE-D3 bulk water dataset,⁵⁵ which consists of 604 training and 50 test configurations of 64 water molecules. The experimental value of $\epsilon_\infty = 1.78$ for bulk water was assumed in the BEC analysis using Eq. (5).

Figure 2(a) compares the test force RMSEs for the short-range baseline models (hollow bars) and their LES-augmented counterparts (solid bars) across several MLIP architectures. LES systematically reduces the force errors for all models, although the magnitude of the improvement depends on the underlying short-range architecture.

Beyond improved energies and forces, LES-enabled models recover physically meaningful electrical response properties. As shown in Ref. 61, all LES-augmented MLIPs accurately predict BECs, indicating that the long-range electrostatics are captured in a physically consistent manner. As a representative example, Fig. 2(b) shows the close agreement between the DFT BECs and those predicted by the CACE LES model trained only on energy and forces (E + F). Importantly, the inference of both long-range electrostatics and BECs within LES is also highly data efficient. As shown in Fig. 2(c), the learning curves for the MACE LES model trained only on the energy and forces of different numbers of configurations demonstrate that accurate BEC predictions emerge rapidly and converge even faster than the forces. The BECs are not only sensitive indicators of the correct electrostatics but also are useful for computing electrical response properties, such as the water IR spectra shown in Fig. 2(f).

Finally, Fig. 2(d) compares the inference speed of the short-range baselines and the LES-augmented models for MACE and CACE. The results show that the additional computational cost associated with the long-range correction is small relative to the baseline evaluation cost, confirming that LES provides an efficient route to incorporating long-range electrostatics without sacrificing performance or system size scalability.

The performance of LES, achieved with a simple algorithm and implementation, is surprising to us: long-range electrostatics for MLIPs turns out to be easier than we thought. In this Perspective, we distill the underlying design principles that make LES work, discuss other possible variants and extensions that also satisfy these principles, highlight the limitations and open questions, and outline our outlook for the future employment of long-range MLIPs.

II. TWO SIMPLE PRINCIPLES

We argue that two simple design principles, outlined in the following, are sufficient for the easy and reliable incorporation of long-range electrostatics into MLIPs under the LES framework.

A. Principle 1: Coulomb functional form with environment-dependent charges

This principle ensures the learned long-range interaction is physically grounded: the one-dimensional environment-dependent charge on each atom is physically interpretable, and the Coulomb form Eqs. (1) and (2) guarantees the correct asymptotic behavior and may help extrapolate beyond trained charge separation distances.

Several long-range models shown in the blue region of Fig. 1, including 4G-HDNNP,²³ PhysNet,²² BAMBOO,²⁵ and AIMNET2,³² also satisfy this principle. In contrast, models in the purple region of Fig. 1 do not explicitly rely on physical charges. For example, although LODE^{39,40} encodes Coulomb and other asymptotically decaying potentials ($1/r^p$) in the atomic environment, its high-dimensional features cannot be directly interpreted as physical charges. Similarly, Ewald MP³⁸ learns long-range descriptors in reciprocal space using trainable frequency filters, without explicitly enforcing a $1/r$ interaction form.

It is difficult to assert which strategy will yield better training accuracy: physically constrained electrostatics or fully learned long-range features. In principle, more flexible learned formulations can capture a broader class of long-range interactions, including dispersion,^{38,40} and may, therefore, achieve lower fitting errors. However, Coulomb forces represent the dominant contribution among long-range interactions, while most others are much weaker. At the same time, all long-range interactions are intrinsically weak relative to short-range forces, yet involve many more atoms, making them fundamentally harder to learn from finite training datasets without inductive bias. The empirical comparisons of these models on specific datasets are scarce in the literature. Fuchs *et al.*⁴⁸ reported that an Allegro⁶⁹ model with a charge equilibration layer achieves performance comparable with that of a PaiNN model⁷⁵ with Ewald MP³⁸ on the OE62 dataset.⁷⁶ Several models, including 4G-HDNNP,²³ charge-constrained ACE,³³ LES-augmented models,^{59,61} LOREM,⁴⁴ CELLI⁴⁸ that uses a charge equilibration layer, charge-equilibrated TensorNet (QET),⁷⁷ and NequIP-LR,⁴⁹ have been benchmarked on some of the datasets [$\text{C}_{10}\text{H}_2/\text{C}_{10}\text{H}_3^+$, $\text{Ag}_3^{+/-}$, $\text{Na}_8/\text{Cl}_8^+$, and Au_2 on $\text{MgO}(001)$] compiled by Ko *et al.*²³ Among these, LES and LOREM have some of the lowest test energy and force errors. However, it has been questioned whether these datasets cover very challenging long-range interactions.⁴⁴ Moreover, these existing comparisons often use different short-range MLIP baselines, making it unclear whether the difference in accuracy stems from the baseline itself or the long-range correction.

Here, we test how the dimensionality of the long-range features affects learning accuracy. In the original formulation of LES,²¹ the latent charge q^{les} can, in principle, be multi-dimensional, with the total long-range energy obtained by aggregating the contributions from different dimensions after the Ewald summation. Conceptually, such multi-dimensional latent charges are analogous to the high-dimensional long-range features used in Ewald MP³⁸ and LODE.^{39,40} As shown in Fig. 2(e), CACE LES models trained with more latent charge output dimensions exhibit essentially negligible and inconsistent improvements in accuracy. While there are alternative ways to exploit high-dimensional long-range information, such as equivariant features,⁴⁴ this example demonstrates that increasing

the dimensionality of long-range features does not necessarily lead to higher training accuracy.

In our view, the greatest advantage of adhering to an explicit Coulomb interaction form is interpretability. This interpretability extends beyond the conceptual meaning of “charges” and directly enables the computation of physically measurable quantities. As discussed in Refs. 59–61, the learned latent charges are predictive of molecular dipoles and higher-order multipoles and also enable the modeling of the electrical response properties such as IR spectra, ionic electrical conductivities, and ferroelectricity. Moreover, the charges can be useful in hybrid schemes such as coupling quantum mechanical calculations with MLIPs (QM/ML)⁷⁸ or combining MLIPs with classical force fields with fixed charges (ML/MM).⁷⁹

B. Principle 2: Not explicitly train on DFT charges

There is no unique way to map the continuous DFT electron density onto discrete partial charges: different partitioning schemes,^{24,80,81} such as Mulliken,⁸² Hirshfeld,⁸³ and MBIS,⁸⁴ often yield substantially different results. Consequently, training on one specific definition of DFT charges would imply a different long-range electrostatic description than training on another. In addition to the uniqueness issue, a more subtle point is that the Coulomb interactions between atomic charges are screened by the rapidly responding background electrons with the relative permittivity ϵ_∞ .^{85,86} This screening is naturally absorbed in the LES formulation through learning from long-range electrostatic forces,⁶⁰ but would generally be absent if atomic charges were learned directly. Consistent with this view, explicit training on DFT charges has also been shown to induce pathological behavior, leading to energy and force errors significantly larger than those of the short-range baseline,²³ such that an additional charge equilibration scheme was required to mitigate such issues.²³

Avoiding training on DFT charges also offers a practical advantage. Most MLIP training datasets, especially those originally designed for short-range models, provide only atomic positions, energies, forces, and sometimes stress tensors. By not requiring charge labels, long-range augmentation becomes immediately applicable to a much broader range of existing datasets and systems. It is worth mentioning that the LES architecture allows training on dipoles and/or BECs, which are physical quantities that can help improve the training.

Moreover, explicit training on DFT charges does not appear to improve energy and force accuracy compared to LES. For the dataset of a diatomic gold cluster Au₂ supported on the MgO(001) surface,²³ which includes both wetting and non-wetting adsorption geometries and where aluminum dopants deep beneath the surface strongly affect the electronic structure and relative stability, a CACE⁷⁰ model augmented with LES and trained only on energies and forces⁵⁹ achieves nearly an order-of-magnitude lower errors than both the charge-constrained ACE model^{33,87} and the 4G-HDNNP.²³

More importantly, we argue that the LES charges learned just from energies and forces are physically meaningful: they correspond to effective partial charges governing electrostatic interactions and are predictive of measurable physical observables. As shown in Refs. 59 and 61, the learned latent charges yield accurate molecular dipoles and multipoles and can further derive BECs using Eqs. (4) and (5).⁶⁰

For the water example, Fig. 2(b) demonstrates that the LES BECs trained on energy and forces (E + F) are in excellent agreement with the DFT reference. These BECs enable the calculation of electrical response properties, such as IR spectra and electrical conductivities.⁶⁰ Figure 2(f) shows the corresponding IR for bulk water: both the spectral shape and intensities are in good agreement with the experiment, capturing distinct high-frequency intramolecular and low-frequency intermolecular modes.

Finally, with the predicted BECs, one can compute the electrostatic forces on atoms under an applied external electric field and perform molecular dynamics simulations under finite fields using MLIPs. Using this capability, we have shown that LES-augmented MLIPs can reproduce electric-field-dependent phenomena, including the IR spectral shift of bulk water under static fields, the ionic conductivity of a superionic material, and the polarization–electric field hysteresis loop of a ferroelectric material.⁶⁰

III. OTHER POSSIBLE EXTENSIONS

Once the two design principles are met, LES can be combined with various additional strategies to suit specific applications. Here, we present several examples.

A. Universal compatibility with any short-range MLIPs

LES can serve as a universal augmentation to any short-range MLIP⁶¹ and we have demonstrated its integration with methods such as MACE,¹² NequIP,¹¹ Allegro,⁶⁹ CACE,⁷⁰ CHGNet,⁷¹ and UMA.¹⁴ Among these, MACE, NequIP, CACE, CHGNet, and UMA use node features to predict local atomic energy contributions and the same features are straightforwardly used to predict the latent charges. Instead, Allegro uses edge features to predict pairwise energies, which are then summed to obtain per-atom energies; in the same spirit, the LES augmentation with Allegro predicts the pairwise charge contributions first using the same edge features and then aggregates them to form the per-atom latent charges.

B. Optional fine-tuning using dipoles or BECs

Although energies and forces alone are sufficient to recover the correct electrostatics, additional training labels, such as dipoles or BEC tensors, can be incorporated for fine-tuning. For example, we show that fine-tuning CACE LES models using DFT-computed BECs further improves BEC prediction accuracy while preserving energy and force accuracy for both bulk water and PbTiO₃ datasets.⁶⁰ The water example shown in the middle panel (E + F + BEC) of Fig. 2(b) illustrates this modest improvement in BEC predictions. As expected, the corresponding IR spectrum shown in Fig. 2(f) remains in good agreement with the experiment.

C. Integrating optional charge equilibration (Qeq)

Empirically, we have found charge equilibration schemes to be unnecessary within the LES framework.^{21,59} For neutral systems, the sum of the learned charges is typically very close to zero; any small residual is treated as a uniform background charge, corresponding to the tinfoil boundary condition in the reciprocal-space evaluation of the electrostatic interactions in Eq. (2).

Nevertheless, LES can be straightforwardly combined with charge equilibration using the Qeq approach,^{47,53} which determines the atomic charges q_i by minimizing the energy,

$$E_{\text{Qeq}} = E_{\text{elec}} + \sum_{i=1}^N \left(\chi_i q_i + \frac{1}{2} J_i q_i^2 \right), \quad (6)$$

where E_{elec} is the Coulomb interaction between charges, χ_i values represent the atomic electronegativities, and J_i values represent element-specific hardness parameters. The χ_i values are predicted by a neural network from local invariant features, while J_i is assigned based on element type. Consistent with the LES design principles, no DFT charges are used for training.

We implemented LES combined with Qeq (LES-Qeq in Fig. 1), with trainable χ_i and J_i . For the water example, the right panel (E + F + Qeq) of Fig. 2(b) shows that the addition of Qeq preserves the accurate predictions of forces and LES BECs. The corresponding predicted IR spectrum shown in Fig. 2(f) also exhibits good agreement with the reference. However, the computational cost of Qeq is substantial: the CACE LES-Qeq model in the right panel of Fig. 2(d) shows that incorporating Qeq leads to cubic scaling due to the matrix inversion operation involved. As such, methods have recently been proposed to reduce this cost, including using the iterative solution of Qeq instead of direct matrix inversion⁸⁸ and developing a particle mesh iterative solver.⁸⁹

D. Coupling with the Siepmann-Sprik metal model

As illustrated by the LES-MetalWall in Fig. 1, the LES can be coupled with the Siepmann-Sprik metal model.⁹⁰ LES determines the charges on the electrolyte, and the Siepmann-Sprik method describes the response charges on metal electrodes. This hybrid long-range framework enables nanosecond finite-field MD simulations of electrode-electrolyte interfaces under periodic boundary conditions.⁷²

E. Charge and spin embedding

For datasets containing multiple charge or spin states, a degeneracy problem can arise: configurations that are geometrically identical but electronically distinct share the same representations in standard MLIPs, which can lead to poor model performance.^{23,33} To lift the degeneracy, one can embed the charge and/or spin states directly in the local atomic environment features. This can be achieved either by encoding global charge or spin states to inform the local features of all atoms,^{59,91} or by utilizing atomwise states such as the oxidation state or the magnetic moment for each atom.⁹¹ As such, not only will the predicted atomic energies depend on these states but also the latent charges.

F. Learning atomic dipoles or dispersion coefficients

The expression for the long-range energy in Eqs. (1) and (2) can be straightforwardly generalized to other $1/r^p$ interactions, such as charge-dipole ($p = 2$), dipole-dipole ($p = 3$), or dispersion-like ($p = 6$) terms. In these cases, the learned latent variables would no longer be scalar charges but higher-order quantities: for example, equivariant atomic dipole vectors for dipolar interactions or per-atom dispersion coefficients for van der Waals-type contributions.

IV. LIMITATIONS AND OPEN QUESTIONS

There are known limitations of the LES algorithm: First, the latent atomic charges q^{les} are determined from local features, and thus, do not provide an explicit mechanism for truly long-range charge transfer mediated by mobile carriers. In materials with metallic or highly delocalized electronic character, an applied external field can induce charge rearrangements that are, in principle, global. An example here is the induced surface charge on a macroscopic metal electrode. In such cases, a purely local LES scheme is not sufficient to describe global charge equilibration. A possible remedy is to couple LES to an explicit metallic boundary model, as in LES-MetalWall, where LES controls the electrolyte charges and the Siepmann-Sprik model performs charge equilibration on the metal electrodes.⁷²

Second, extracting BEC tensors for periodic systems via Eq. (5) requires specifying a constant high-frequency dielectric permittivity ϵ_∞ for a homogeneous bulk material.⁶⁰ For heterogeneous systems, such as interfaces between materials with different ϵ_∞ , it is not obvious how to choose or generalize this parameter. Developing consistent strategies to extend LES-based BEC extraction and response-property calculations to inhomogeneous or interfacial systems remains an open challenge.

Third, scalability to very large system sizes remains an engineering challenge. The nominal $\mathcal{O}(N^{3/2})$ scaling of Ewald summation itself is not the main bottleneck, as this operation is typically inexpensive relative to other components of modern MLIPs and more favorable scaling variants, such as particle-particle particle-mesh (P3M) or particle-mesh Ewald (PME), are readily available.⁴³ Rather, the main difficulty arises because interatomic forces are obtained via the automatic differentiation of the total energy with respect to atomic positions, where the total energy is a global quantity that includes the long-range electrostatic contribution, which makes it nontrivial to distribute force evaluation across multiple GPUs.

Finally, there is currently no “foundation model”⁹² that spans broad regions of chemical and materials space while incorporating LES. Encouragingly, MACELES-OFF,⁶¹ trained on the SPICE dataset⁹³ of molecules and small clusters, is more accurate than its short-range counterpart MACE-OFF,⁹⁴ predicts dipoles and BECs reliably, and provides improved descriptions of bulk liquids. However, it remains an open question whether LES augmentation will offer similar advantages for substantially larger, more diverse molecular datasets or bulk materials datasets.^{95,96} Exploring this direction is key to determining whether long-range electrostatics can be made standard in next-generation foundation MLIPs.

V. CONCLUSION AND OUTLOOK

In this Perspective, we examine the connections and distinctions between LES and other representative methods for learning long-range interactions (Fig. 1). While many of the aforementioned approaches show great promise, we argue that a minimal and physically grounded design is sufficient for incorporating long-range electrostatics, based on two simple principles: (i) use a physical functional form with flexible, environment-dependent atomic charges to capture electrostatic interactions, and (ii) avoid explicit training on DFT partial charges. In addition, there are many avenues for other optional extensions.

Despite the known limitations, many aspects of the LES method have turned out to be easier than we initially anticipated: the design principles and algorithm are straightforward; only standard energy and force labels are required, training is data-efficient [Fig. 2(c)]; electrical response properties can be extracted reliably [Figs. 2(b) and 2(f)]; and the associated computational overhead is low [Fig. 2(d)].

This naturally raises a broader question: why do short-range MLIPs remain the dominant paradigm, while long-range electrostatic MLIPs have been adopted only sparingly in practical simulations? Equivalently, if long-range electrostatics can, in fact, be incorporated so simply and efficiently, why was this not widely recognized earlier? One plausible reason is that some of the early approaches shown in Fig. 1 do not fully satisfy the two design principles outlined above or may have higher computational expense. Another explanation may be that long-range methods have not yet been pushed and benchmarked extensively enough to convincingly demonstrate their broad applicability and computational efficiency.

Another possibility is simply that the atomistic modeling community has long been used to short-range MLIPs and has not yet made the transition to long-range models. Speaking from our own experience, when one of us (B.C.) began using MLIPs around 2018, it was already striking that atomic interactions could be learned and converged simulations performed with near-quantum mechanical accuracy. At that time, the systems we studied were predominantly homogeneous bulk systems, such as bulk water,⁹⁷ superionic water,⁹⁸ and high-pressure liquid hydrogen.⁹⁹ For these systems, long-range electrostatics is conceptually important, but was not the main concern: long-range interactions are largely isotropic and tend to average out, allowing short-range models to effectively learn the mean-field forces. Many observables of interest, such as radial distribution functions, transport coefficients, and free energies, are not sensitive to long-range electrostatics.¹⁰⁰ At the same time, the training errors of the first-generation MLIPs^{7,8} were substantially larger than those of atomic cluster expansion (ACE)⁸⁷ or MPNNs¹⁰¹ due to the lack of higher-body-order terms, making long-range errors relatively less important. It is only more recently, as our work has shifted toward heterogeneous and interfacial systems, that long-range electrostatics has become essential. Interfaces introduce inhomogeneity, as well as intrinsic polarization and electric fields, where the short-range MLIPs struggle. As the community increasingly studies complex, multi-component, and field-driven systems, especially electrochemical interfaces, polar and ionic materials, and biomolecules, we expect long-range MLIPs to become not just beneficial but necessary.

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AUTHOR DECLARATIONS

Conflict of Interest

B.C. has an equity stake in AIMATX Inc. B.C. has Patent No. 19/307,702 (pending) for the Latent Ewald Summation algorithm.

Author Contributions

Dongjin Kim: Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Bingqing Cheng:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The RPBE-D3 bulk water dataset, training scripts, evaluation scripts, the trained CACE E + F + Qeq model, and CACE LES and MACE LES models used to produce results shown in Figs. 2(c)–2(e) are available at https://github.com/ChengUCB/les_fit.

The LES library is publicly available at <https://github.com/ChengUCB/les>. The CACE package with the LES implementation is available at <https://github.com/BingqingCheng/cace>. The MACE package with the LES implementation is available at <https://github.com/ACEsuit/mace>. The NequIP and Allegro LES extension package is available at <https://github.com/ChengUCB/NequIP-LES>. The MatGL package with the LES implementation is available at <https://github.com/ChengUCB/matgl>. The UMA package with the LES implementation is available at https://github.com/santi921/fairchem/tree/les_branch.

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